Lessons Learned From A Failed Wide-Swing Current Mode R-2R Digital to Analog Converter

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The author investigates critical errors and non-linearity of an implemented Digital to Analog Converter using the Wide-Swing Current Mode R-2R Ladder configuration supplied by a 1.8 V power supply and a stable 900 mV reference voltage, which was to have an output range of 0.4 – 1.4 V and an output clock of 10 MHz. The circuit design did not operate as expected. After thorough diagnostics, it was found that switches were not powered in the R-2R ladder and it led to degeneracy between the input and output. The author was able to successfully create a 0.4 and 1.4 V references independent of supply variation, but suggested improvements to the design could have decreased the area and its dependence on the supply voltage; also, the output clock met the frequency requirements of 10 MHz, but its size was large and its performance could have improved with an adjustment. The system was tested per cell and its mechanisms are described in this piece.

Keywords—Digital to Analog Converter, Wide-Swing Current Mode R-2R Ladder, Beta-Multiplier Reference

I. Introduction

Although the convenience of digital signals have many contemporary electronics commonly using them, there does not exist a digital signal without analogue support, and many of our senses such as hearing cannot interpret these signals; this is an example where digital to analog converters (DAC) may come to use, and even modern Analogue to Digital Converters may use the DAC. The conversions of these DACs have a simple relationship between each bit that differs from its neighboring bit by a factor of two, but design limitations, trade-offs, and parasitic effects will cause non-linear effects such as differential non-linearity (DNL), integral non-linearity (INL), and a shifted offset leading to a bad interpretation of the signal [1].

II. System Overview

An 8-bit DAC was made using an R-2R ladder, capacitor array, or current steering topology. The author had concerns about capacitors in charge scaling being too large to manage given the area requirement of the chip, ≤ 1 mm^2, as it was noted a 1 nF capacitor could account for 0.64 mm^2; conversely, a current-steering topology would have been excellent for the area using MOSFETs alone, but it would have critical matching requirements and would need calibration from process errors in practical applications. It was found that the R-2R ladder offered a reasonable medium in size and matching dependency would not be critical for functionality, but it would have a known effect against linearity. The R-2R also offers an easy to understand contribution for each bit using the superposition principle, and there were several configurations fitting for the target specifications, given inputs, and the requested output.

There were three pronounced configurations of the R-2R: i) the current-mode R-2R, ii) the voltage-mode R-2R, and iii) the wide-swing current-mode R-2R. It was described in the literature that the current-mode R-2R would lead to issues with the dynamic range by giving up half of its voltage and placing more demand on the operational amplifier, and the voltage-mode R-2R would have linearity issues because of the variable off-set voltage; the wide-swing current-mode R-2R offered an easy configuration for the output range dependent on the switches into the ladder and into an inverted operational amplifier [1]. This topology had a parallel input requirement although the designer was given a serial input; so: A serial to parallel converter (SIPO) would be required. The author was also provided with a power supply of 1.8 V, *VDD*, and a stable voltage reference of 900 mV, *VREF*, but would require voltage references for the output range between 400 mV and 1.4 V. Lastly, a 10 MHz internal clock was requested. The resulting overview would match fig. 1.

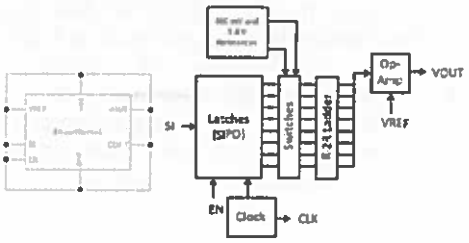


Fig 1. Symbol and design overview

The reference voltage would be of large concern asa simple voltage divider would have too much supply dependency; the author had wrongfully assumed that customers could use large capacitors at the supply nodes as suggested in most other datasheets, which led to failed preliminary design review and a heavy focus steered towards the reference voltages. The author was also incorrectly told during the preliminary design review that transmission gates should not be used for the switches, but the literature suggests otherwise in chapters [2], examples, and the output range could be easily be resolved by the feedback resistor and an offset in the inverted op-amp [1].

III. Circuit Implementation

A. Latches/SIPO

Latches for the SIPO worked perfectly using D-type flip flops (DFF) in series. The design of the DFF used transmission gates, which preserved the solid "high" and "low" values even at rates faster than requested, and this was proven in simulation with inputs stressed to 2 GHz shown in figure 1.

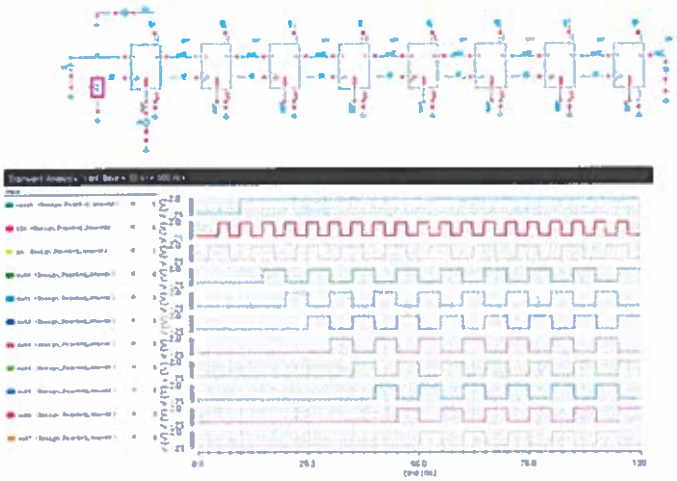


Fig. 2. Schematic of the SIPO with its transient waveform.

B. Voltage References

The DAC needed a stable 400 mV and 1.4 V voltage references, V_{REF+} and V_{REF-} respectively, to connect to the switches within the ladder; these would more easily set the output range. There were several good options for voltage references such as band-gap voltage reference circuit [1-2, 4], but concerns were not related to temperature as the specifications had a constant 27 °C ambiance temperature. The other options with voltages dividers using either diode connect transistors or resistors had too much dependency on *VDD*. The beta multiplier reference (BMR) was a clear choice of the candidates, and a standard configuration was used for this design [2,4]. The transistors were coscoded to meet higher voltages, which were roughly scaled by the threshold voltage of the transistors [4] and the final design is shown in fig. 2.

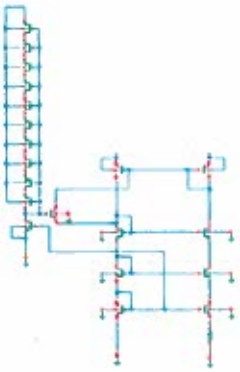


Fig. 3. Schematic of the BMR and start-up circuit.

It was understood that cascoding would also increasing output resistance and place more requirement on the start-up circuit to overcome these threshold requirements. Increasing reference resistance would decrease the dependency on supply voltage [1], which is shown in fig. 3. The author had chosen arbitrarily and based on practice to reduce this variation to roughly 20% of the target output voltages, but the trade-offs of less dependency increased the area of the design.

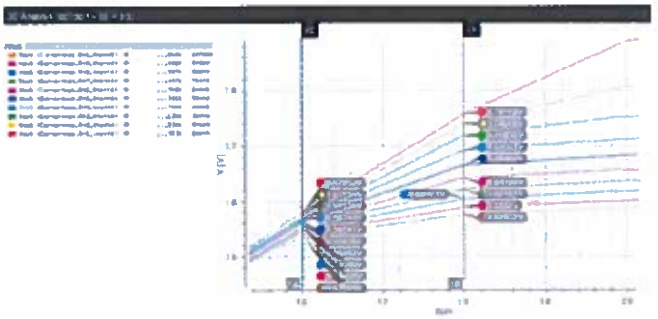


Fig. 4. Simulations of the BMR confirming supply dependency were proportional to the reference resistance.

C. Switches

The switches used two inverters in a series. The first inverter was powered by *VDD* to ground, but the following inverter was supplied by V_{REF+} and V_{REF-} from the BMR circuit. It could be seen that a high bit would output V_{REF+} and a low bit would output V_{REF-}, and the results of the simulation is shown in fig. 4. It is commonly known that the switches have a rise and fall time, but the rise and fall times of inverters were set to less than 200 ns and would not interfere with maximum frequency requirement of 500 kHz. The outputs of these switches were fed directly into the R-2R ladder. Although simulations of the switches clearly worked as expected, the submitted version used aliases to connect the nodes, which caused the switches to not have a power supply at all and convert input signals into ringing output. The ringing would cause degeneracy because all outputs would look the same regardless of input. At the time of submission, the author's suspicion went directly towards the operational amplifier as ringing output suggests instability. These non-supplied inverters were the most dominating issue of the entire system; otherwise: The DAC would have had been in working order and calibration could have corrected the remaining issues.

D. R-2R Ladder

The submitted version of the R-2R ladder was incorrectly configured with the *R* and *2R* swapped; however, the ladder takes advantage of the superposition principle and hand calculations could show what each bit would contribute. Unfortunately, a swapped *R* and *2R* would not distinguish each bit by a factor of two. Most interestingly, the minimum and maximum output voltage would not change, but everything in between would not be linear. Another noticeable issue with swapped *R* and *2R* resistors is that the output would ring in transient simulations, but this effect was not prominent until after 2 ~ 3 µs. The submitted design also should have tied the

least significant bit (LSB) to V_{REF-} instead of the ground, and this issue affected the linearity of the ladder heavily overreaching the desired output range. With *R* and *2R* resistors having been swapped, this incorrect grounding made the least LSB to have an unexpected contribution to the output. The value of *R* heavily affects the power dissipation and needed to increase to reduce it, which can be "hand waived" using $P = V^2/R$ where *R* is inversely proportional to power. The resistance *R* was set to 1 kΩ, but this was too low because the output current was 5 μA, which caused design to exceed allowable dissipation power. Although the R-2R has less critical matching requirements, mismatches between the resistors increases the DNL as seen in equation (1); therefore: Dummy resistors and a common centroid layout should be used.

$$DNL = \Delta I \cdot R \approx (V_{REF+} - V_{REF-}) \cdot [(\Delta R/R) - 2^{-N}] \quad (1)$$

After issues with the switches were resolved, the output voltage was still outside of range at 700 mV – 1.2 V even with perfect sources from a generic analog library. This suggested that calibration would have to be done using the operational amplifier.

E. Operational Amplifier

The operational amplifier was the most frequency dependent component. It is known that op-amp could become unstable and oscillate, or at least have a cut-off at higher frequencies causing the user to have a significantly different gain than expected. This design was very dependent on having an inverted op-amp configuration with the common mode connected to V_{REF} and the feedback resistor would could off-set the remaining linearity issues in the design as shown by the following equation:

$$V_{OUT} = V_{REF-} + [(V_{REF+} - V_{REF-})/2] + I_F \cdot R \quad (2)$$

The original design of the operational amplifier had a cut-off frequency that was too low because Virtuoso has known issues when using both fingers and multiplier not accepted into Virtuoso, which has simulations shown in fig. 5. The author had to redesign the operational amplifier to have larger transistors until the 3 dB drop occurred at ~ 170 kHz with simulated results shown in fig. 6; this redesign led to issues with the layout as the op-amp has several critically matched components such as a current mirror and differential pair, and it consumed the most time during the project.

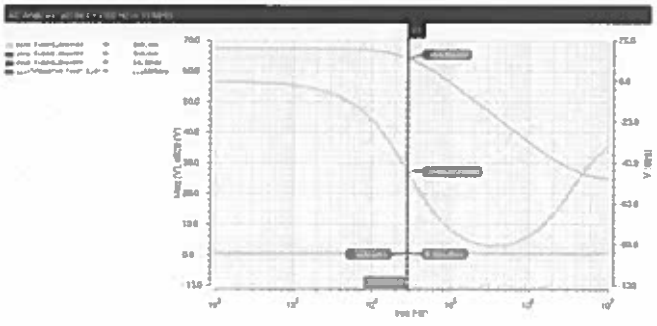


Fig. 5. Op-amp simulation shown to have low cut-off frequency at 29 kHz.

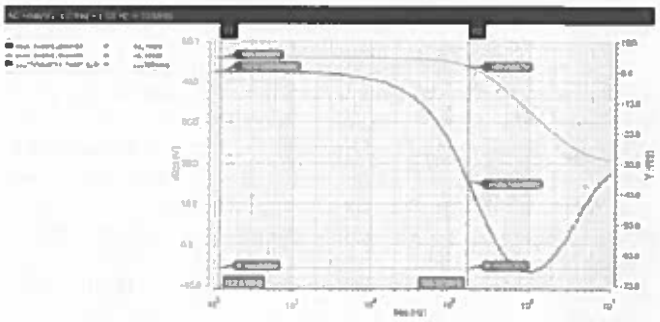


Fig. 6. Op-amp simulation with improved cut-off frequency at nearly 170 kHz.

F. Ring Oscillator

The ring oscillator used five stages of inverters that were fed back into its input, and two stages of inverters as a buffer output. The initial designs with the pmos and nmos having a width of 2 μm and 1 μm would not meet the frequency requirements, and the maximum output voltage was not large enough to indicate a high for other cells such as the DFF in the SIPO; it was also requested that the lengths be at least three times their minimum length of 180 nm (designated by the gdpk library) for good practice. Little information about the library could be found so far as the capacitance of the oxide and so a parametric sweep of the widths and lengths had to be used to determine fitting values for the requested oscillations. The author found without the use of resistor and capacitor components, that the 10 MHz requirement could be met with lengths set to 5.5 μm and the pmos and nmos set to widths of 72 μm and 36 μm respectively; an attempt with resistors and capacitors was made, but it was clear that these components would heavily dampen the oscillation even at small magnitudes, and it would be difficult to meet frequency requirements. The resulting ring oscillator is shown to have 10 MHz frequency with a period roughly at 100 ns in fig. 7.

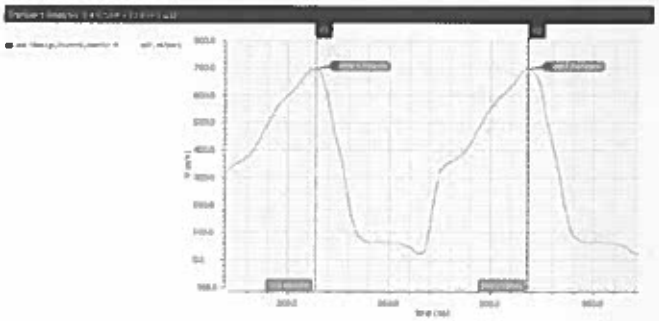


Fig. 7. Ring Oscillator having nearly 100 ns period indicating the required 10 MHz frequency.

It is noticed that the high and the lows are not very strong, and the author believes that a Schmitt trigger can set these values if desired, but it did not appear necessary for the functionality of the circuit.

IV. Conclusion

The submitted design did not work as expected, but this was because of the switches within the R-2R not having been supplied with power; having recognized this issue, the submitted design would had linearity issue due to a swapped *R* and *2R* resistors in the ladder and it did not meet the power requirements because these resistor values were too low in an effort to reduce size. The author attempted a working version having resolved these problems and has shown in the mechanisms above the feasibility of the design. A BMR circuit was created, which gave a reference very independent of the voltage supply, and previous versions of the operational amplifier and ring oscillator were improved to meet the target specifications. It has been further shown in the literature that this topology would likely perform better in practice compared to simulated results [1].

References

[1] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulations, 3rd ed., IEEE Press: Piscataway, 2010, pp. 612-652.

[2] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulations, 3rd ed., IEEE Press: Piscataway, 2010, pp. 745-770.

[3] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulations, 3rd ed., IEEE Press: Piscataway, 2010, pp. 829-833.

[4] C. Toumazou, F. J. Lidgey and D. G. Haigh, Analogue IC design: the current-mode approach, IEE Circuits and Systems Series 2, Vol. 1, Short Run Press, Ltd.: Exeter, 1998, pp. 481-487.

Coursework Assessment

Full Custom Design of a Digital to Analogue Converter (DAC) in 0.18μm CMOS

jkp16

Jan, 2017

Your DAC is not functional as expected

1 Overall Mark

Your Overall Design Mark is F
Your Overall Report Mark is D

2 Design Specifications

Table 1: Achieved Design Specifications

Name	Specifications	Value	Units	Rank	Best	Median
Area	Total silicon area	N.A.	μm^2	90% - 100%	6300.00	43056.00
P_{Sdiss}	Static Power dissipation	6.200	mW	50% - 60%	1.75	5.67
P_{Ddiss}	Dynamic Power dissipation	4.424	mW	20% - 30%	1.67	5.17
ERR_{rel}	Relative error	233.660	%	90 % - 100%	0.15	0.55
ERR_{clk}	Clock error	13.318	%	90 % - 100%	0.00	0.42

Best is the best results across 41 students and Median is the median value

3 Comments On The Design

Please find comments for the design submission as below

3.1 Design and functionality

R-2R DAC, also designed beta cell for reference. Resistive voltage divider provides poor output drive ability for the DAC, also using INV instead of Switch could suffer from speed issue.

3.2 Design clarity

In top level, transistors and resistor can be grouped into new or exsit cell for clarity.

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3.3 Layout

Your DAC layout did not pass DRC, Please note this may be due to lack of design files or imcomplete designs
Your DAC layout did not LVS, Please note this may be due to lack of design files or imcomplete designs

Layout is not finished

4 Report Evaluation

Please find comments for your report as below

4.1 Design description

Lots of descritpion of the design blocks but some key parts are missing such as the schematics of the R-2R network. Design errors are not marked here but a few things to keep in mind are: switchingthe reference through an inverter and driving through a beta multiplier means that the current has a long path leading to a very slow response. Buffering through an opamp would fix this in your design. Also TGs can and should be used to implement switches - there was definitely a misunderstanding at your preliminary design review. Ringing seems like an opamp problem. Also regarding the use of multipliers, Virtuoso considers it as circuits appearing in parallel therefore the names of the nets around it affect how the circuit is actually designed. As a result the op amp that you designed was probably different than what you thought and it is not a softwarre issues. The output of the ring oscilaltor should be a square wave - a Schmitt trigger is not needed just a buffer (inverter) receiving the output from the ring oscillator. When sizing the rign oscillator you don't need any resistors or capacitors, just sizing of each inverter to derive the necessary frequency. Also, you mention that the lengths need to be at least 3 times the minimum length - this only concers analogue circuitry and is not applicable here, what more digital than a clock? In any case, minimum length wouldn't help you here since the size of the gate WxL determines its capacitance and thus your oscillation frequency. Moreover, the resistor values of 1K should have worked there must have been a different problem with the R-2R network. Overall you spend a lot of time for advanced concepts such as supply-independent references, wide-swing current-mode R-2R etc whereas much simpler solutions would have probably been finished by the deadline.

4.2 Circuit Theory

Some theory present but a lot of issues not properly understood regaring circuit operation.

4.3 Results and interpretation

Some simulation results shown but not of complete operation.

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4.4 Presentation

Most figures are illegible. References are not enough are adequate and should include a book once.

5 Simulation Log

Please find simulation log as below, please note these were the initial simulation results, and were reviewed by markers. Manual resimulated might be performed to get accurate specification for some points

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This is evaluation results of Full Custom IC Design course work for jkp16
Creating test bench schematics (tcellDACforscript_jkp16 in CMOS16_DAC_JKP16) for design (DAC in CMOS16_DAC_JKP16)
Found terminal vdd of inst INST0
Matched to pin vdd! in the assignment
Found terminal gnd of inst INST0
Matched to pin gnd! in the assignment
Found terminal CLK of inst INST0
Matched to pin CLK in the assignment
Found terminal VOUT of inst INST0
Matched to pin VOUT in the assignment
Found terminal EN of inst INST0
Matched to pin EN in the assignment
Found terminal SI of inst INST0
Matched to pin SI in the assignment
Found terminal VREF of inst INST0
Matched to pin VREF in the assignment
CLK period is 113.3175 nS, relative error is 13.3175 Static power consumption is 6.1999 mW , dynamic power consumption is 4.4237 mW
Index Reference Vout error
0 0.4039 1.3477 2.3366
1 0.4078 1.3478 2.3048
2 0.4118 1.3479 2.2736
3 0.4157 1.3481 2.2430
4 0.4196 1.3482 2.2130
5 0.4235 1.3483 2.1836
6 0.4275 1.3485 2.1547
7 0.4314 1.3486 2.1263
8 0.4353 1.3488 2.0985
9 0.4392 1.3489 2.0711
10 0.4431 1.3490 2.0443

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11 0.4471 1.3492 2.0179
12 0.4510 1.3493 1.9920
13 0.4549 1.3495 1.9665
14 0.4588 1.3496 1.9415
15 0.4627 1.3498 1.9168
16 0.4667 1.3499 1.8926
17 0.4706 1.3500 1.8688
18 0.4745 1.3502 1.8454
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20 0.4824 1.3505 1.7997
21 0.4863 1.3506 1.7775
22 0.4902 1.3507 1.7555
23 0.4941 1.3509 1.7339
24 0.4980 1.3510 1.7127
25 0.5020 1.3512 1.6918
26 0.5059 1.3513 1.6712
27 0.5098 1.3514 1.6509
28 0.5137 1.3516 1.6309
29 0.5176 1.3517 1.6112
30 0.5216 1.3518 1.5918
31 0.5255 1.3520 1.5727
32 0.5294 1.3521 1.5539
33 0.5333 1.3522 1.5354
34 0.5373 1.3523 1.5171
35 0.5412 1.3524 1.4990
36 0.5451 1.3525 1.4813
37 0.5490 1.3526 1.4637
38 0.5529 1.3527 1.4464
39 0.5569 1.3528 1.4294
40 0.5608 1.3529 1.4125
41 0.5647 1.3530 1.3959
42 0.5686 1.3531 1.3795
43 0.5725 1.3531 1.3634
44 0.5765 1.3532 1.3474
45 0.5804 1.3533 1.3316
46 0.5843 1.3533 1.3161
47 0.5882 1.3534 1.3007
48 0.5922 1.3534 1.2856
49 0.5961 1.3535 1.2706
50 0.6000 1.3535 1.2558
51 0.6039 1.3535 1.2412
52 0.6078 1.3536 1.2268

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53 0.6118 1.3536 1.2126
54 0.6157 1.3536 1.1986
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79 0.7137 1.3538 0.8969
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85 0.7373 1.3538 0.8363
86 0.7412 1.3538 0.8266
87 0.7451 1.3538 0.8170
88 0.7490 1.3538 0.8075
89 0.7529 1.3538 0.7981
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91 0.7608 1.3539 0.7795
92 0.7647 1.3539 0.7704
93 0.7686 1.3539 0.7614
94 0.7725 1.3539 0.7524

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95 0.7765 1.3539 0.7436
96 0.7804 1.3539 0.7348
97 0.7843 1.3539 0.7262
98 0.7882 1.3539 0.7176
99 0.7922 1.3539 0.7091
100 0.7961 1.3539 0.7007
101 0.8000 1.3539 0.6923
102 0.8039 1.3539 0.6841
103 0.8078 1.3539 0.6759
104 0.8118 1.3539 0.6678
105 0.8157 1.3539 0.6598
106 0.8196 1.3539 0.6518
107 0.8235 1.3539 0.6440
108 0.8275 1.3539 0.6362
109 0.8314 1.3539 0.6285
110 0.8353 1.3539 0.6208
111 0.8392 1.3539 0.6132
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114 0.8510 1.3539 0.5909
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119 0.8706 1.3539 0.5551
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121 0.8784 1.3539 0.5412
122 0.8824 1.3539 0.5344
123 0.8863 1.3539 0.5276
124 0.8902 1.3539 0.5208
125 0.8941 1.3539 0.5142
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127 0.9020 1.3539 0.5010
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137 0.9412 1.3539 0.4385
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177 1.0980 1.3539 0.2330
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179 1.1059 1.3539 0.2242
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182 1.1176 1.3539 0.2113
183 1.1216 1.3539 0.2071
184 1.1255 1.3539 0.2029
185 1.1294 1.3539 0.1987
186 1.1333 1.3539 0.1946
187 1.1373 1.3539 0.1905
188 1.1412 1.3539 0.1864
189 1.1451 1.3539 0.1823
190 1.1490 1.3539 0.1783
191 1.1529 1.3539 0.1743
192 1.1569 1.3539 0.1703
193 1.1608 1.3539 0.1663
194 1.1647 1.3539 0.1624
195 1.1686 1.3539 0.1585
196 1.1725 1.3539 0.1546
197 1.1765 1.3539 0.1508
198 1.1804 1.3539 0.1470
199 1.1843 1.3539 0.1432
200 1.1882 1.3539 0.1394
201 1.1922 1.3539 0.1356
202 1.1961 1.3539 0.1319
203 1.2000 1.3539 0.1282
204 1.2039 1.3539 0.1245
205 1.2078 1.3539 0.1209
206 1.2118 1.3539 0.1173
207 1.2157 1.3539 0.1137
208 1.2196 1.3539 0.1101
209 1.2235 1.3539 0.1065
210 1.2275 1.3539 0.1030
211 1.2314 1.3539 0.0995
212 1.2353 1.3539 0.0960
213 1.2392 1.3539 0.0925
214 1.2431 1.3539 0.0891
215 1.2471 1.3539 0.0856
216 1.2510 1.3539 0.0822
217 1.2549 1.3539 0.0789
218 1.2588 1.3539 0.0755
219 1.2627 1.3539 0.0722
220 1.2667 1.3539 0.0688

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221 1.2706 1.3539 0.0655
222 1.2745 1.3539 0.0623
223 1.2784 1.3539 0.0590
224 1.2824 1.3539 0.0558
225 1.2863 1.3539 0.0525
226 1.2902 1.3539 0.0493
227 1.2941 1.3539 0.0462
228 1.2980 1.3539 0.0430
229 1.3020 1.3539 0.0399
230 1.3059 1.3539 0.0367
231 1.3098 1.3539 0.0336
232 1.3137 1.3539 0.0305
Max ERROR is 2.3366
Simulation finished

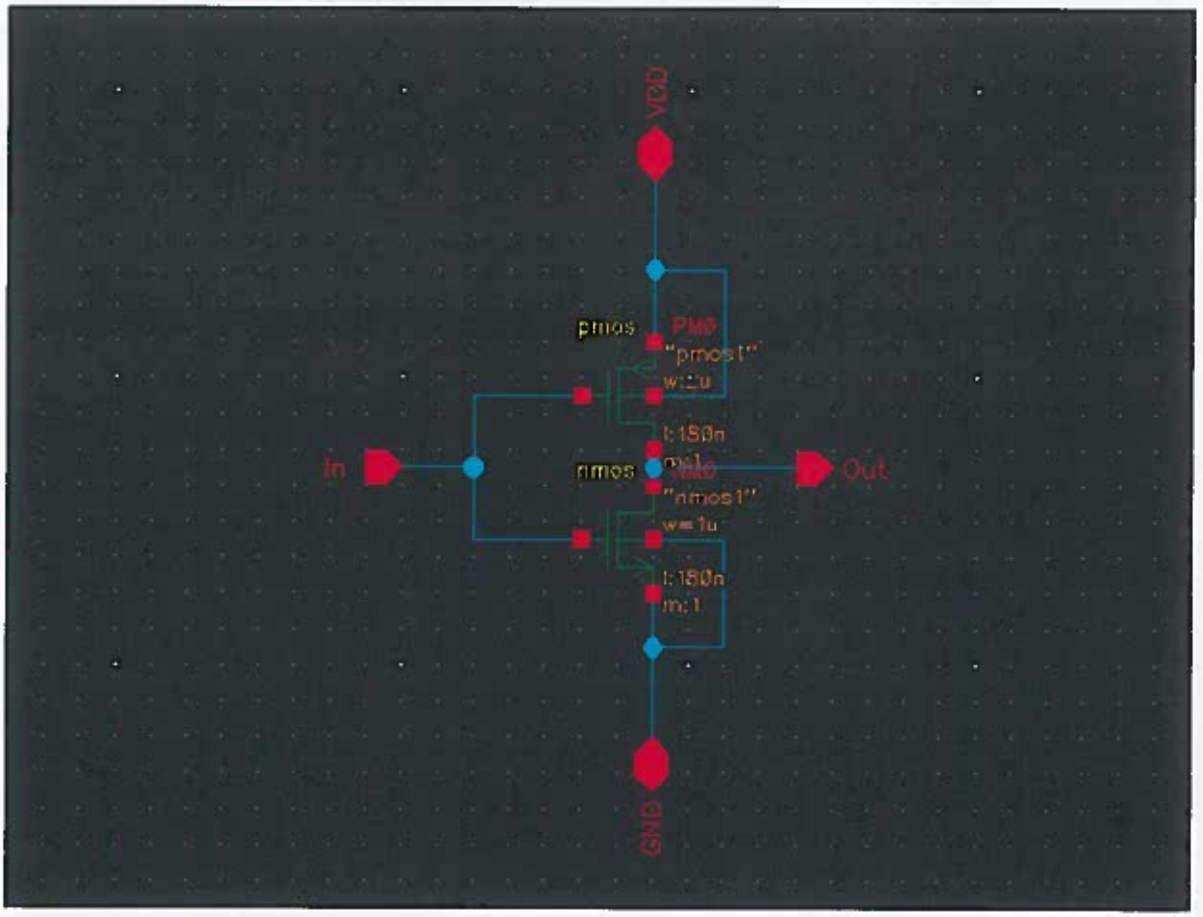
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Joshua Przepioski
CID # 01288971
EE4-20 Full-Custom Integrated Circuit Design: LAB 2

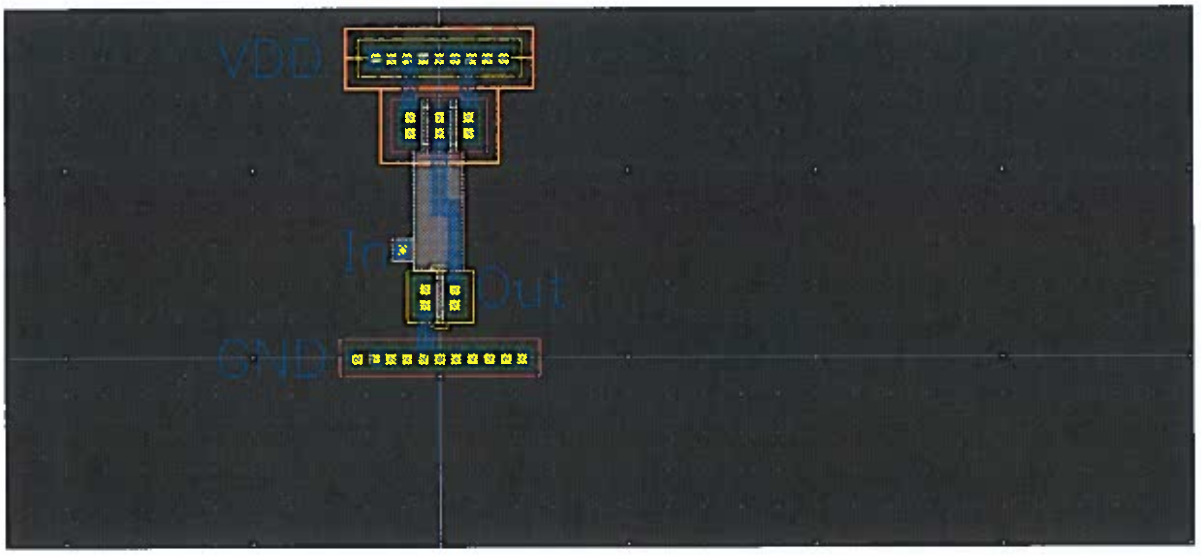
Ultimately: I was not able to have a working DRC and LVS for the D-type Flip Flop because of a failed layout using too much metal two and poor organization. I believe the CLK and its inverter was difficult to find placement for and to connect through other feedback circuits, and I had spent too much effort into understanding the layout for the inverter and NAND gates, which was used to make the D-type Flip Flop.

However: Here is the work I was able to complete successfully. The D-type Flop Flop composed of both Inverters and NAND gates, although the literature [1] had examples using all Inverters or all NAND gates pages 387-388; the one proposed in the manual is the structure I aimed for as to avoid different inputs. This chapter mainly explained the benefit of using a transmission gate (TG) to improve the noise performance compared to Inverters. The layout looks conceptually similar, and they also demonstrate this similarity between the Inverter and NAND gate when you tie the input terminals together. This is what I believe was the strategy for creating they layout, and then you would know the ratio between the PMOS and NMOS sizes should still be roughly two (based on Lab 1 material) to have the correct V_{SP} at 900 mV.

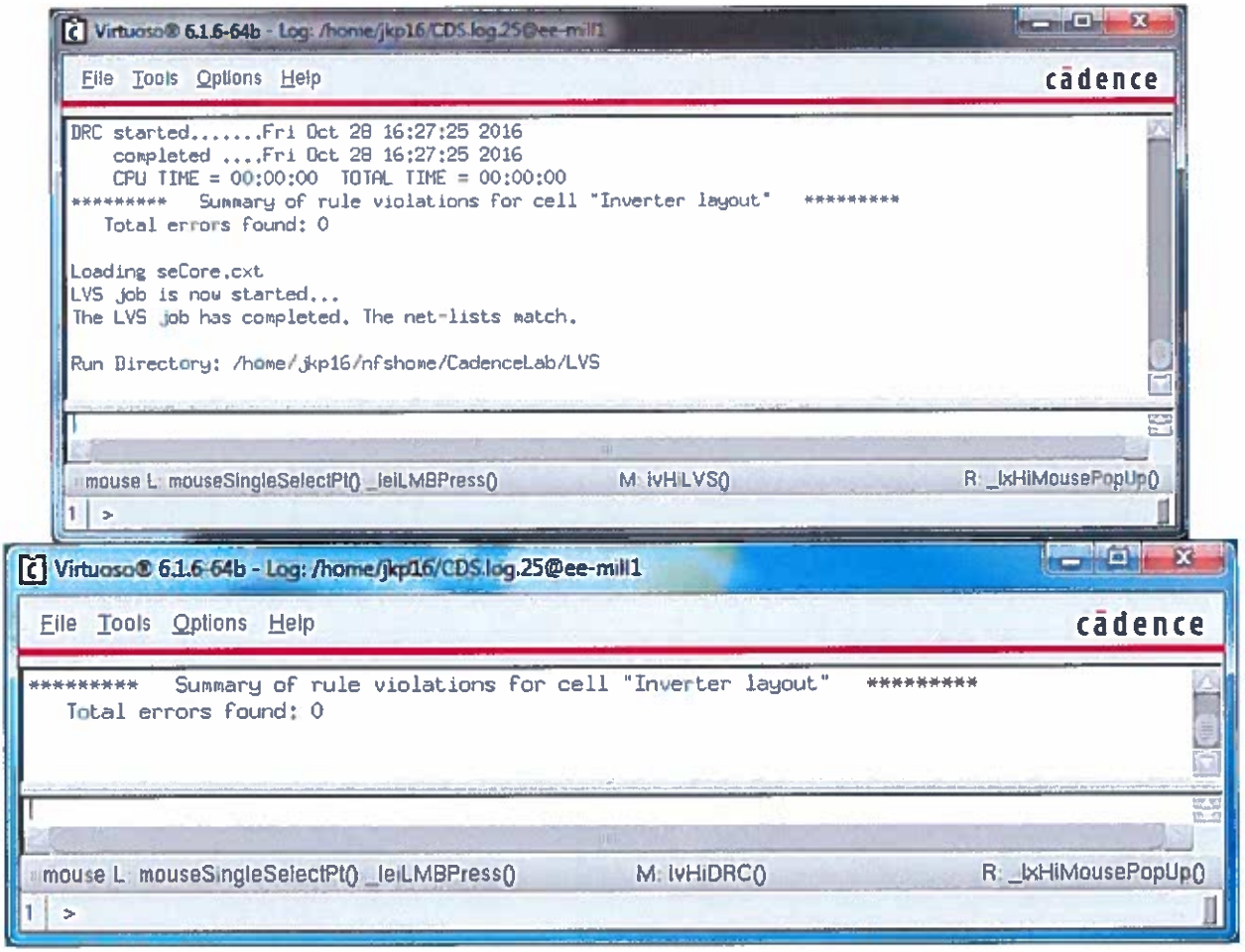
Here is the Inverter I used and demonstrated its functionality in Lab 1.



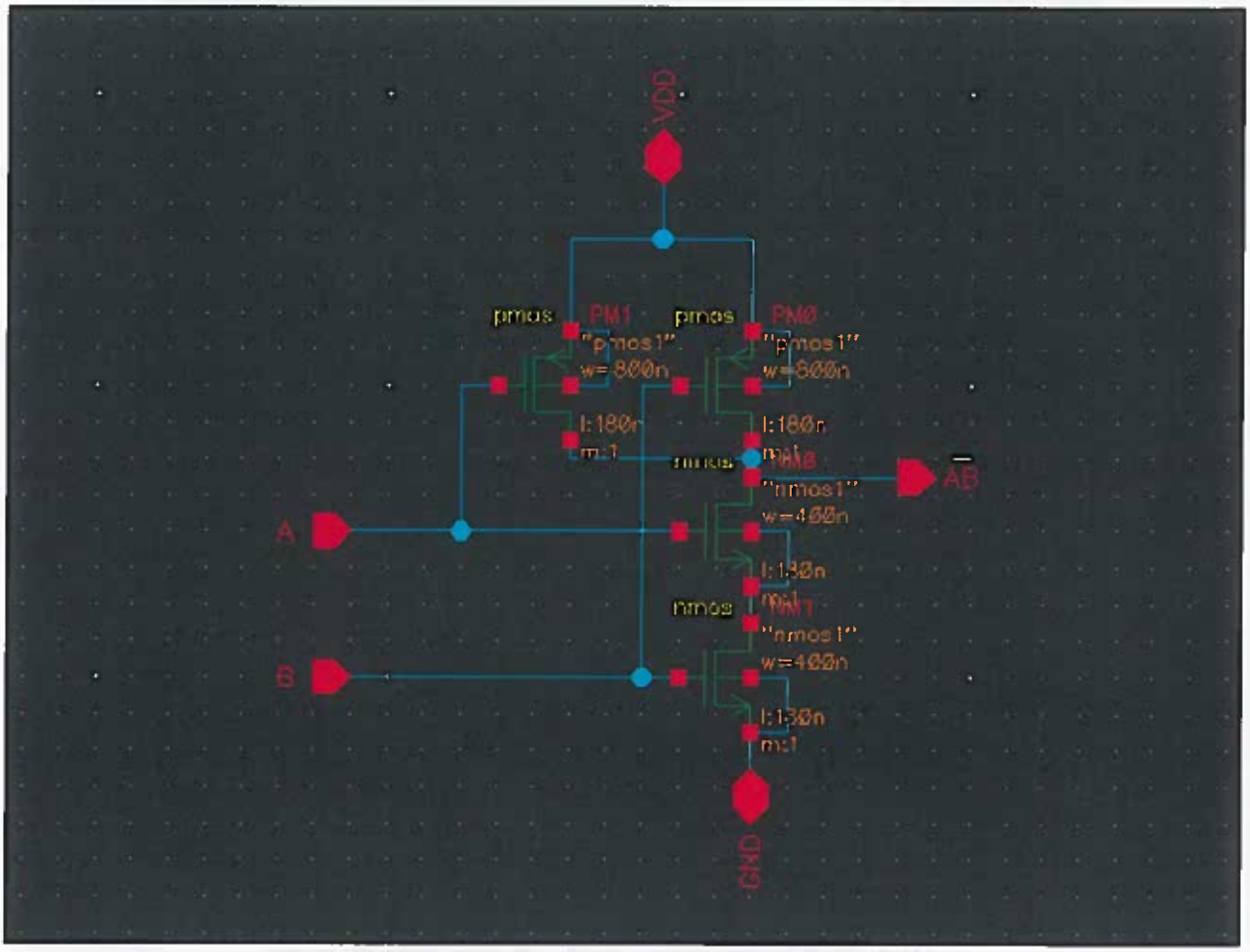
In Lab 2, we created the layout using pretty conventional techniques, although I plan to separate the oxide for spacing to strategically place vias in the future.



I verified that it passed DRC and LVS.



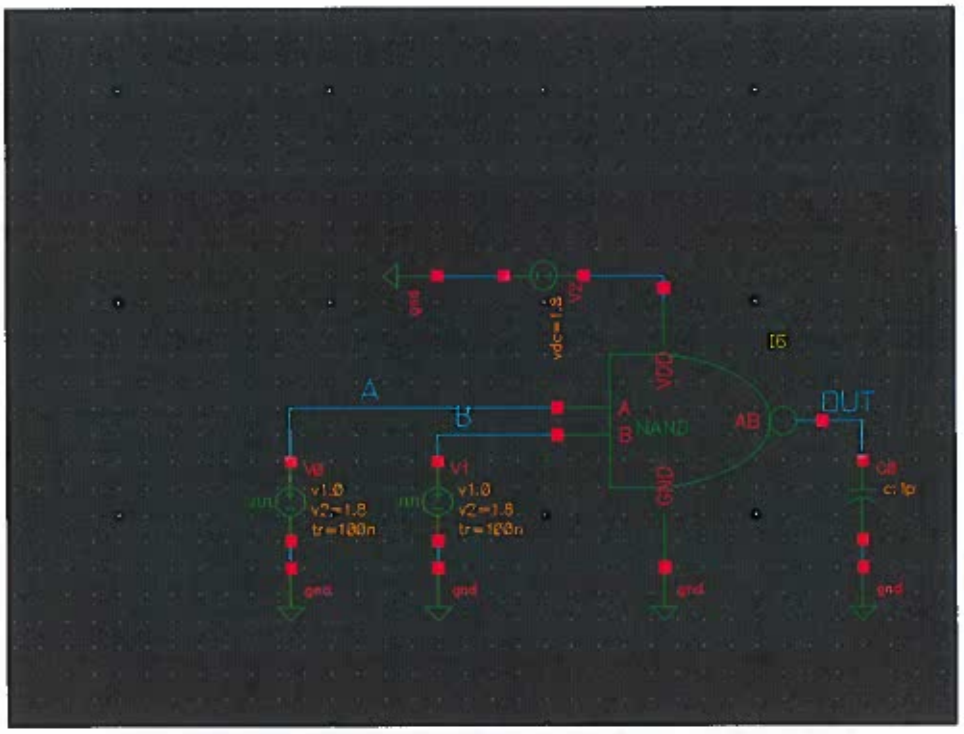
The NAND gate was nothing artistic using a cascade of PMOS and stacking it onto of a cascode of NMOS. Since this was conceptually the same as the Inverter and knowing that you need a VSP at 900 mV, I conserved the rule using twice the width on the PMOS.

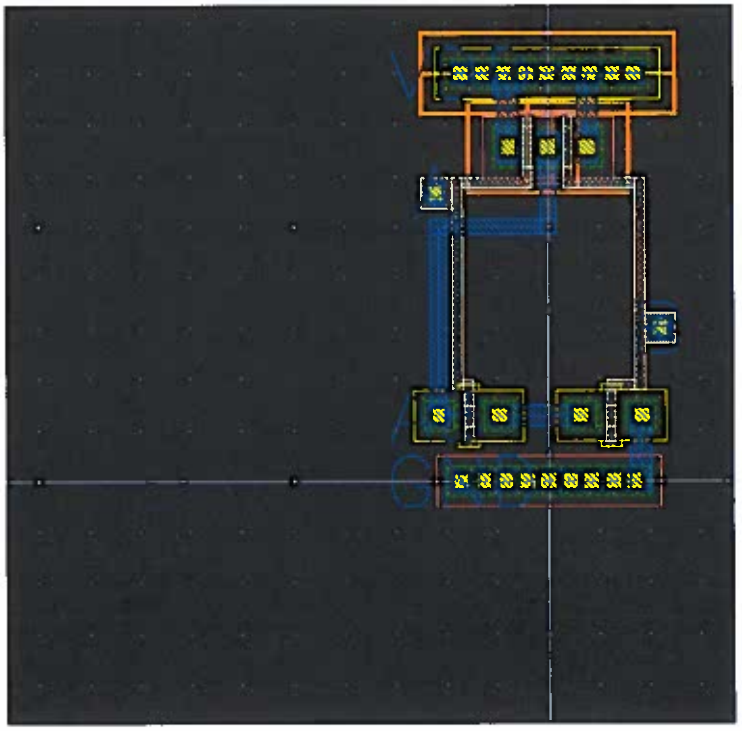


I tested this using two AC sources, one with a period twice as long such that it was low and high for twice as long compared to the other. This made looking at the results very simple. Cadence gave me some struggles in plotting all the graphs simultaneously, but here is the truth table for NAND

A	B	$\overline{AB}$
1	1	0
0	1	1
1	0	1
0	0	1

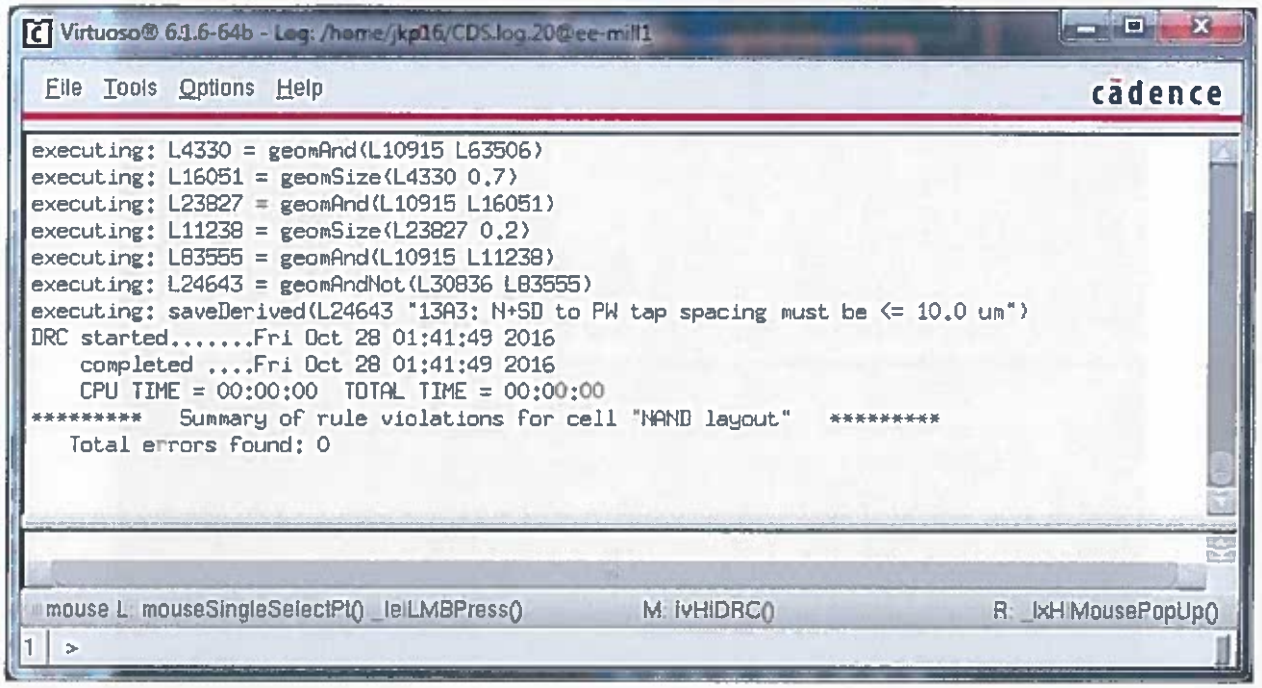
There was a slight dip in near 600 ns in because of the rise and fall time in my testbench, although the graph is attached for verification.



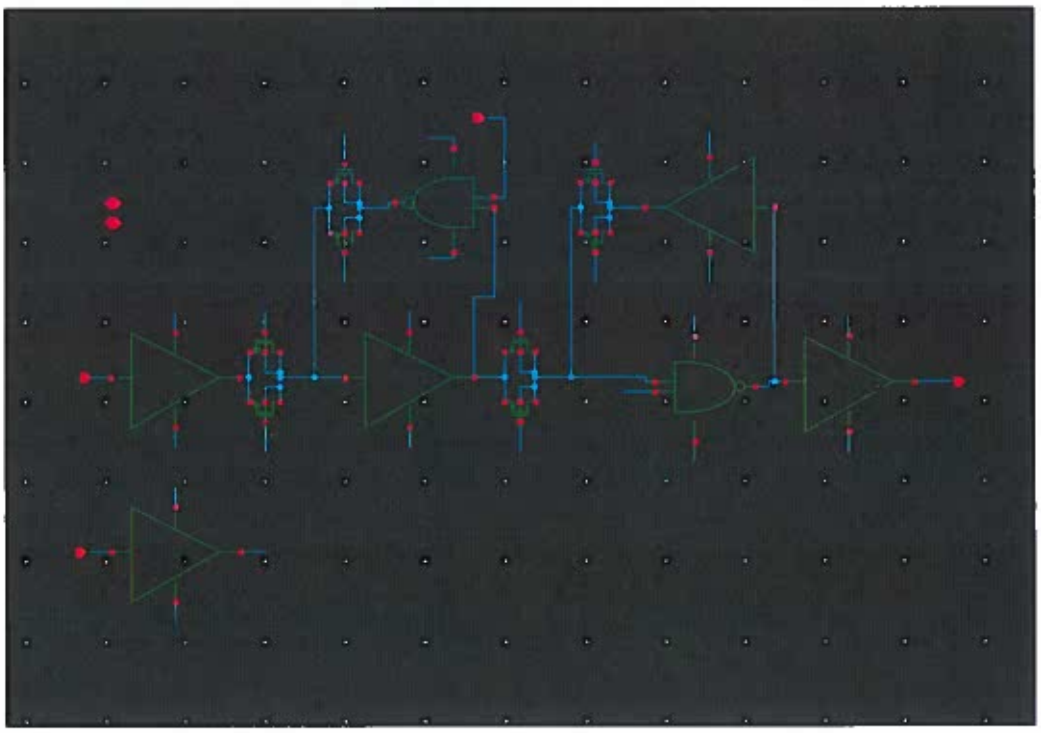


This layout looks very different from the layout shown on page 415 (Fig. 15.5b) in an effort to use the middle region for vias should I need it.

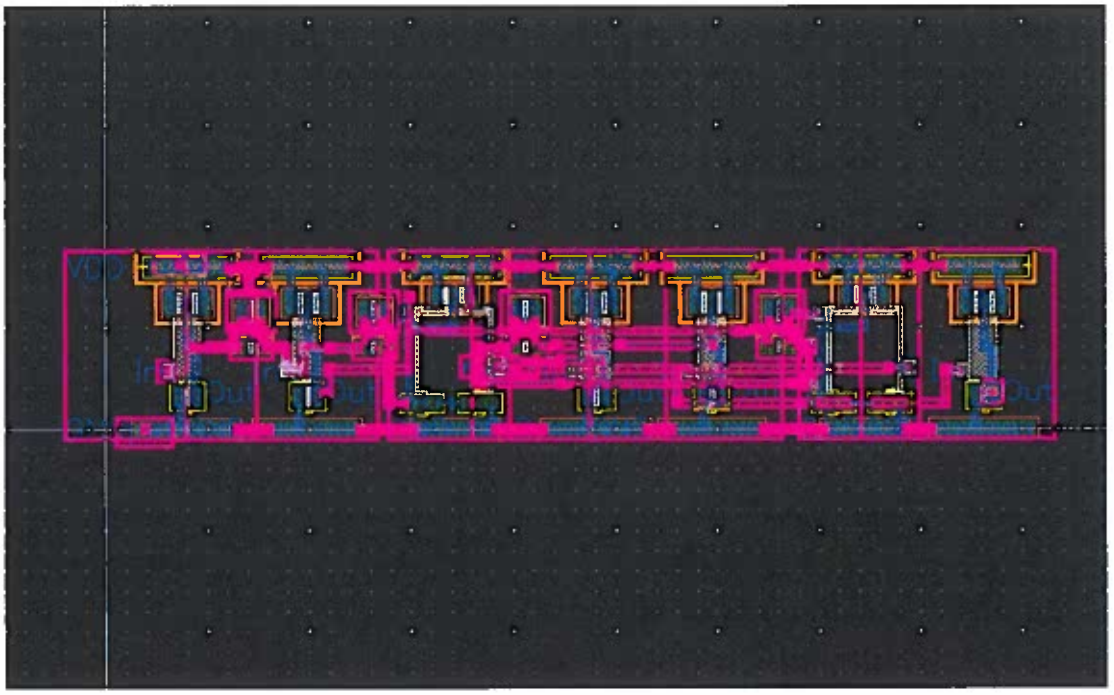
I also confirmed with DRC that it didn't violate rules and LVS for matching nets.



I implemented the D-type Flip Flop layout using the layout from the lab manual.



The errors I came across was in the TGS I had kept misplacing the CLK and $\overline{CLK}$. I had also misplaced the VDD and GND on the pieces that I had rotated; I was aiming to strategically place the CLK inverter in the center because the TGS were spread out rather well and also in a feedback system, which suggested that it had a dominating role in the system.

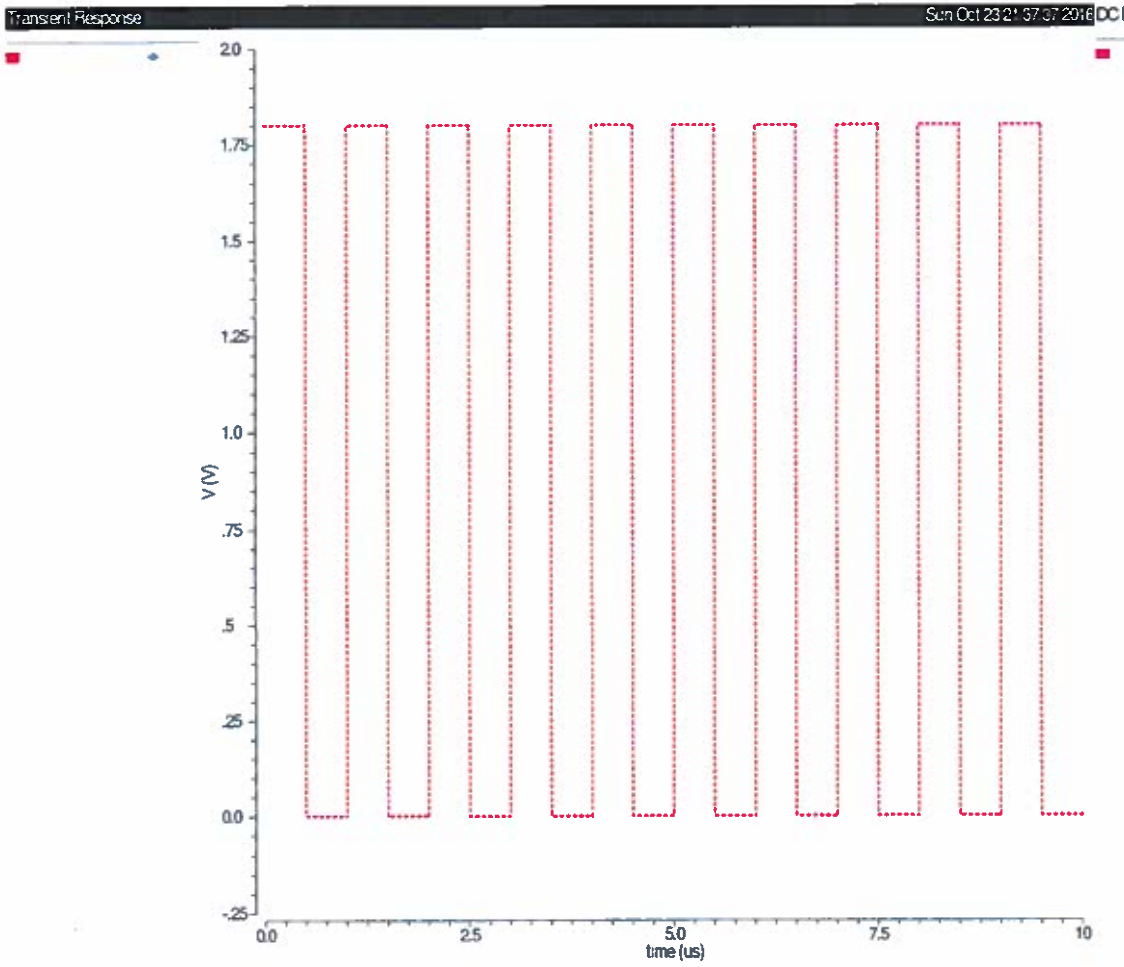


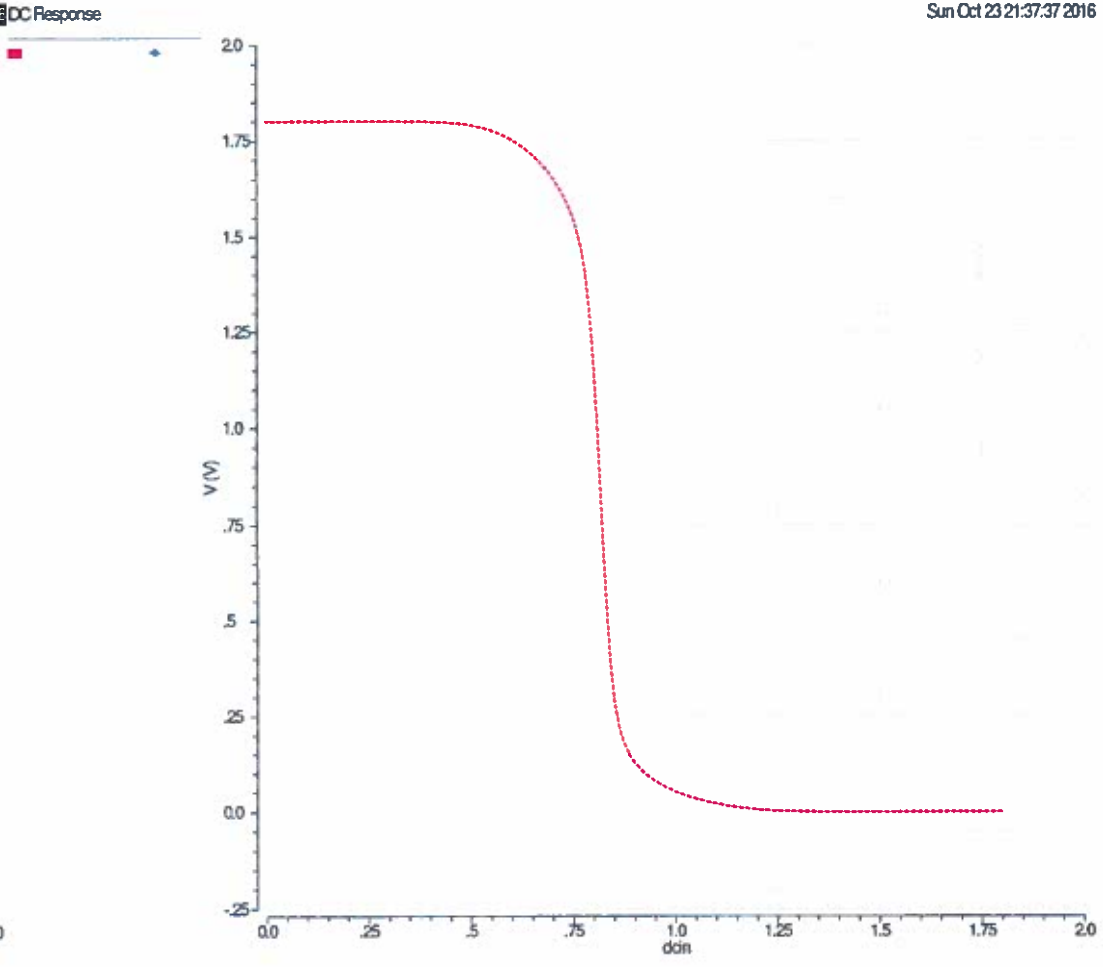
With the inverters having such a large poly silicon space being used, this layout began to rely on metal1

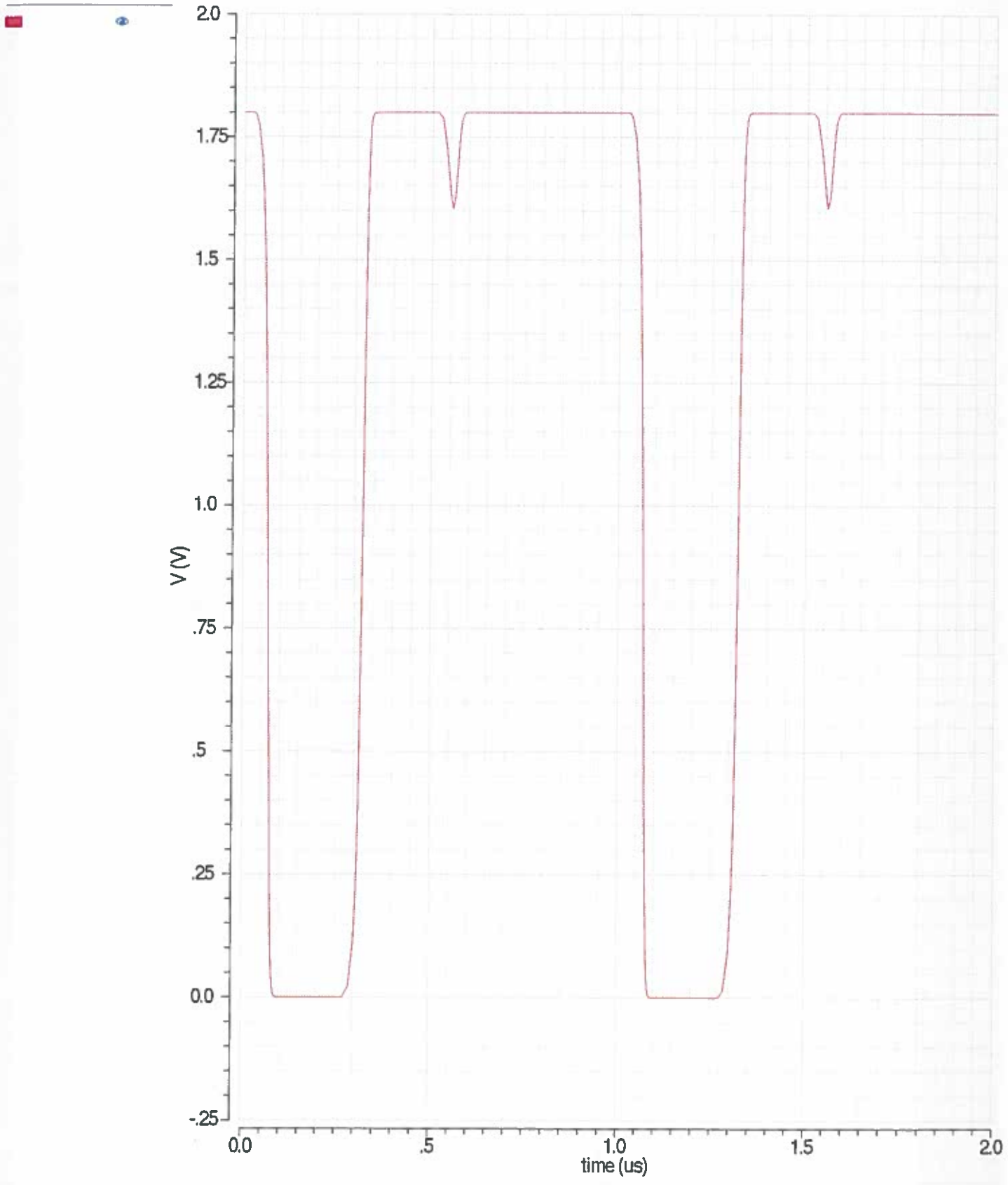
to metal2 connections, which made this a large mess. This was primarily between the clocks, which was coming from the center and reaching both ends of layout. I chose to stay with this method because longer lengths of the route could be interfered by noise and due to time reasons.

I was not able to get a working DRC and LVS.

[1] R. Jacob Baker, CMOS Circuit Design, Layout, and Simulation, Third Edition. Hoboken, NJ: John Wiley & Sons, Inc., 2010, pp.


Transient Response
Sun Oct 23 21:37:37 2016
2.0
1.75
1.5
1.25
1.0
.75
.5
.25
0.0
-.25
V (V)
0.0
2.5
5.0
7.5
10
time (us)


DC Response
Sun Oct 23 21:37:37 2016
2.0
1.75
1.5
1.25
1.0
.75
.5
.25
0.0
-.25
V (V)
0.0
.25
.5
.75
1.0
1.25
1.5
1.75
2.0
dcin


2.0
1.75
1.5
1.25
1.0
.75
.5
.25
0.0
-.25
V (V)
0.0
.5
1.0
1.5
2.0
time (us)

Joshua Przepioski
CID 01288971

SCHEMATIC

In this laboratory, we were given a schematic for a two stage Miller Operation amplifier; the transistors in this schematics make up a differential pair (PM0-1), a current mirror (PM2-5), and NM0-3 transistors are active loads. The schematic provided has transistors that largely vary in width (W) as small as 6 µm, and as large as 100 µm.

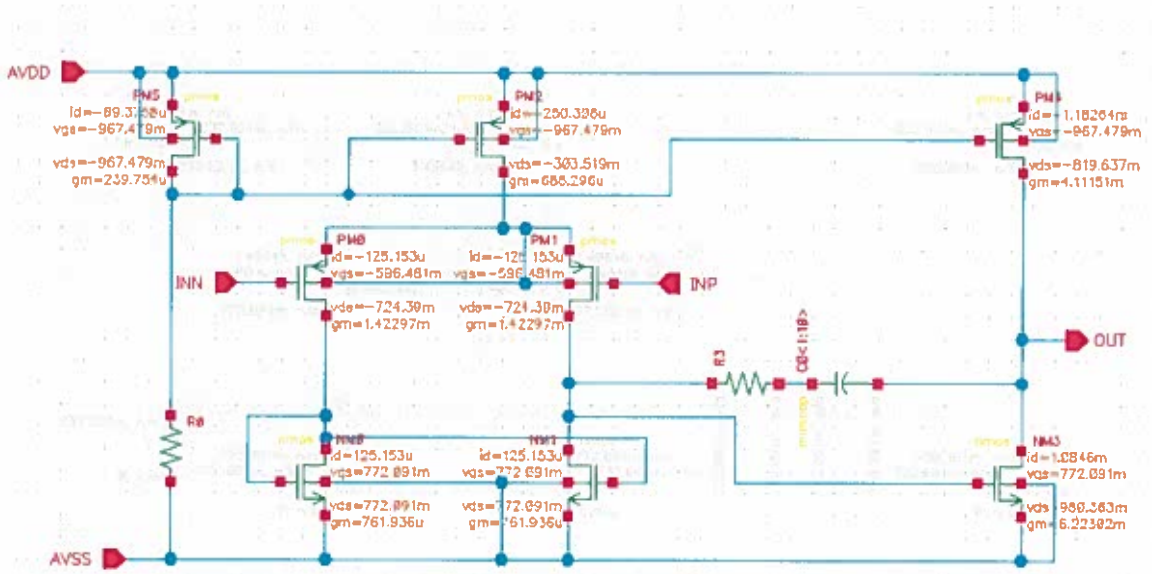


Figure 1 Circuit schematic for op-amp

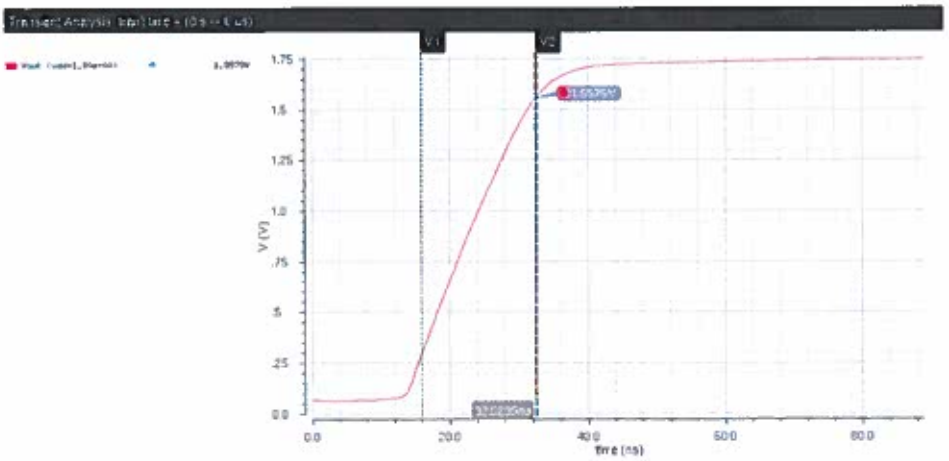
SIMULATION and CHARACTERIZATION

I created a schematic symbol for the op-amp and prepared the test bench schematic with the original parameters described in the lab manual.

Cut-off and Gain

This is where you can see that the gain of the amplifier is shy of 70 dB with its 3 dB drop at $f_{cut\text{-}off} \sim$ 28.6 kHz. The stability of the amplifier is within a very safe range as the phase clearly stays above a 90° drop. What can be done to increase the gain is to vary the resistance between NM3 and PM3, which is where large difference in gain is noticed in later graphs (figure 9 is a case where the gain drops); if you adjust the size of these transistors, the parasitics do not scale perfectly with it.

Slewing



$$slew = \frac{dV_{out}}{dt} = \frac{I_{SS}}{C_L} \approx \frac{1.5575 - 0.292\,V}{32.52 - 16.01\,ns}$$
$$\approx 7.67 \times 10^{7}\,V/s$$

Figure 2 I calculated the slew rate looking closely at the slope of Vout.

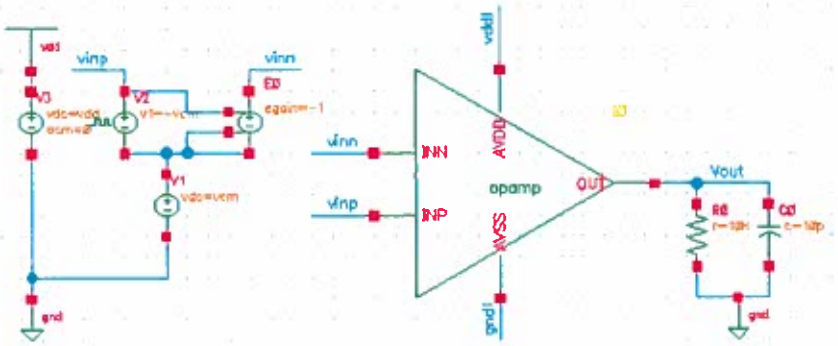


Figure 3 Test bench schematic for op-amp

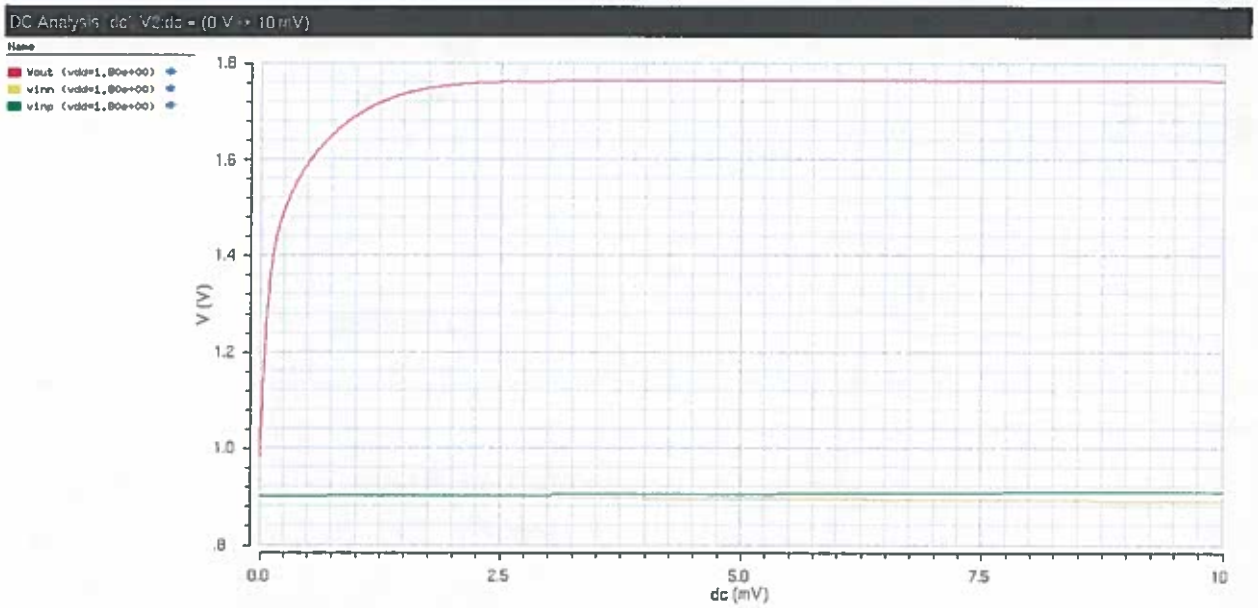


Figure 4 DC sweep input and output results of op-amp. Output voltage levels out at roughly 2.5 mV.

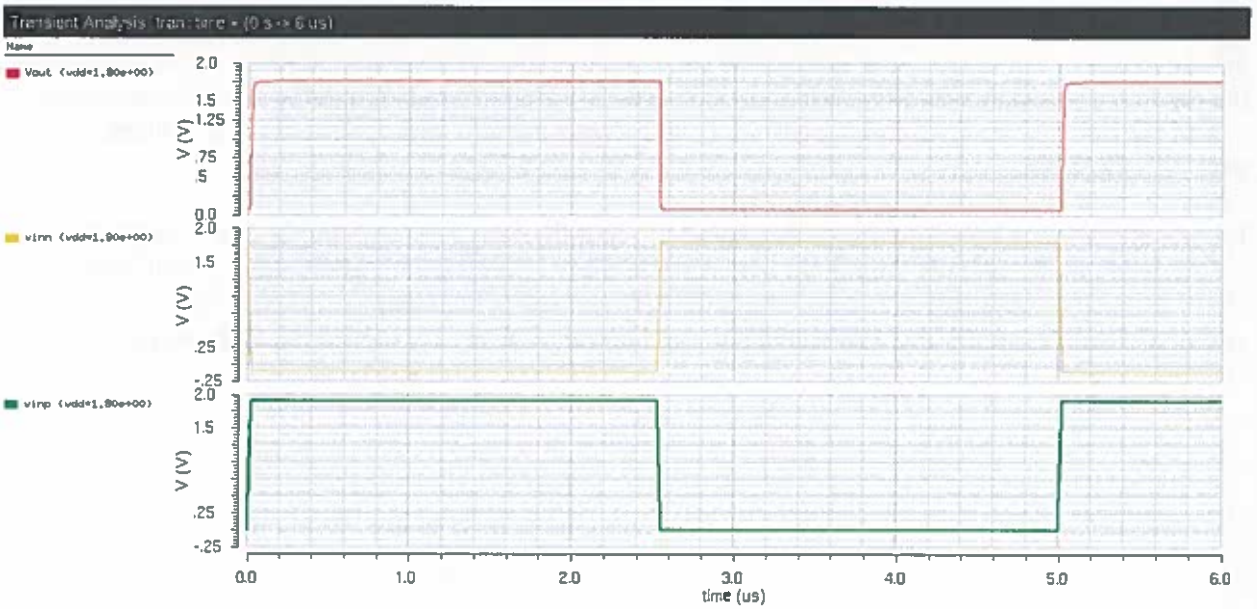


Figure 5 Transient simulation input and output.

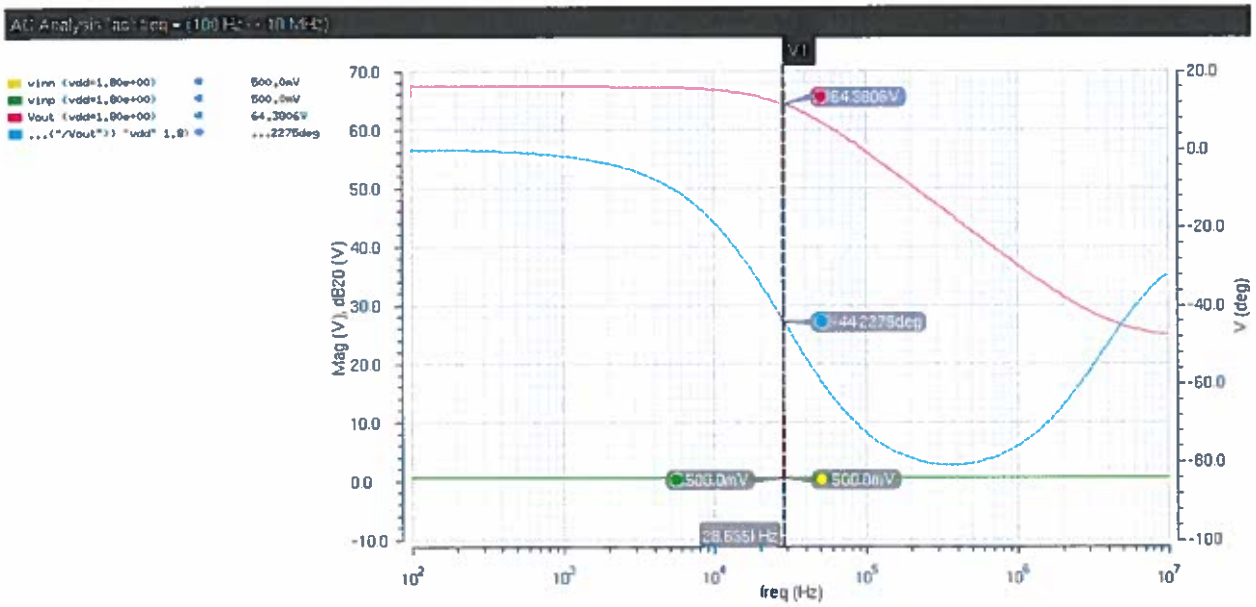


Figure 6 AC simulation of opamp with phase and 3 dB drop

LAYOUT and CONSIDERATIONS

With the large difference of size in the schematics, one would want to do interdigitated layout, but having these current mirrors and differential pairs makes matching a critical factor of its performances; the parasitics (resistance) across each node looks vastly different compared to the other connections, and so we have choice but to use a *common-centroid layout* that is very symmetrical in both the $\hat{x}$ and $\hat{y}$ direction.

I first aimed for any trivial dummy components to see if that may help with the layout in larger sizes. It can be seen that I'm already aiming for symmetry in the $\hat{x}$ direction by adding three components, but then I realized my misunderstanding: The goal isn't symmetry or common-centroid for the greater circuit as a whole, but for the differential pairs and current-mirror. I scrapped this right away.

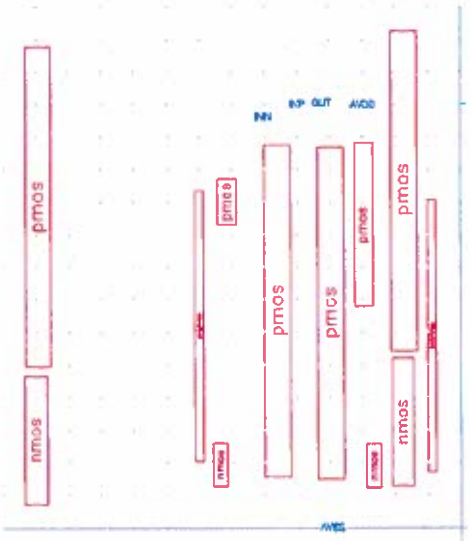


Figure 7 Attempt at using dummy components initially, which would be tied to ground.

I adjusted the transistors so that their widths would be the same, but their $\frac{W}{L}$ factor would change and the current output would not be what I had expected. Another consideration here was using squares since they are very symmetrical; I used this wherever I could. I corrected this by adding multiples of the transistors in parallel to sum their currents together $n \times \frac{W}{L}$.

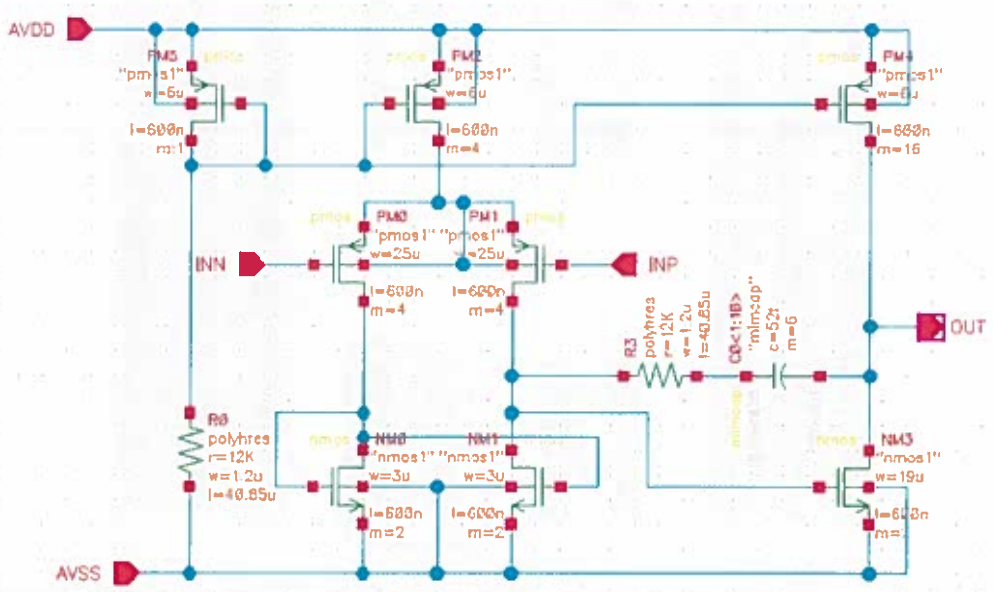


Figure 8 Revising the schematic for more symmetrical components; using smaller "W" but in multiples

Upon revision, the first thing I checked was if any of the characteristics had changed without my knowing. I believe in theory and calculation, these should have been the same results.

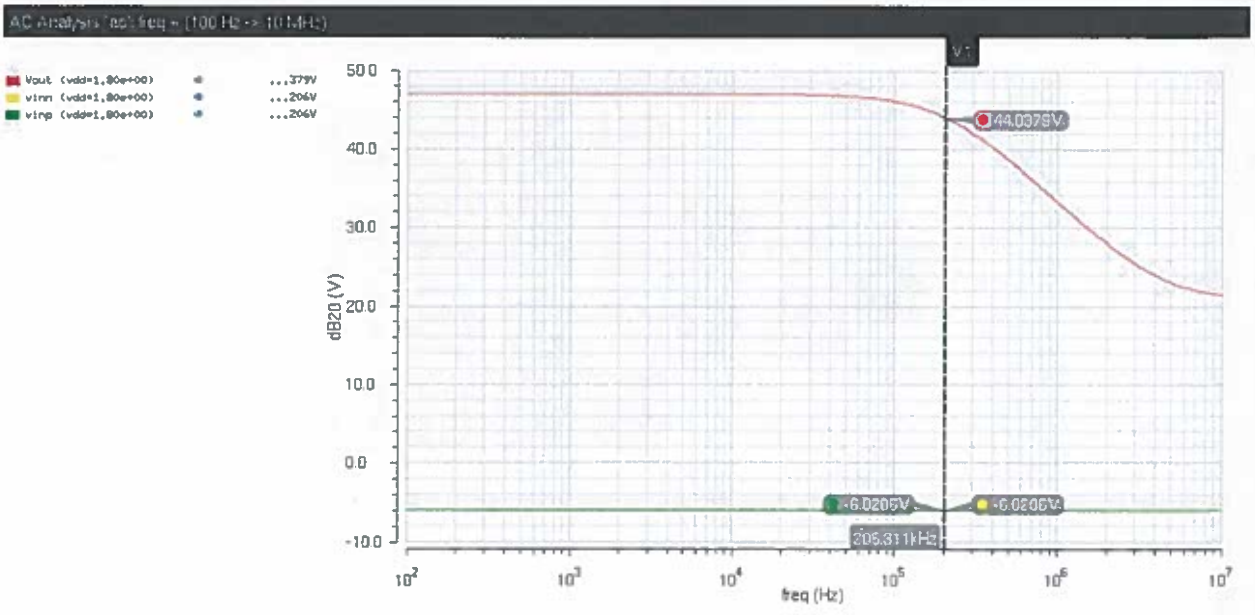


Figure 9 AC simulation after changing W and multiples of transistors

Clearly, this looks like a completely different op-amp because $f_{cut\text{-}off}$ is far different and the Gain has substantially dropped. I noticed that adjusting the transistors at touching either the capacitance (*PM4* and *NM3*) substantially contributed to this effect; also, *PM2* had some effect, but these changes were small relative to these other two (roughly 1 dB).

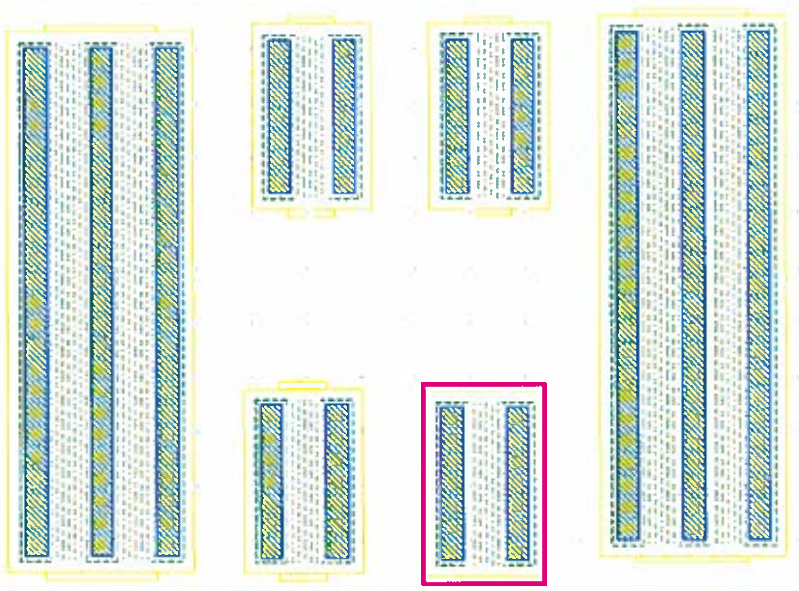
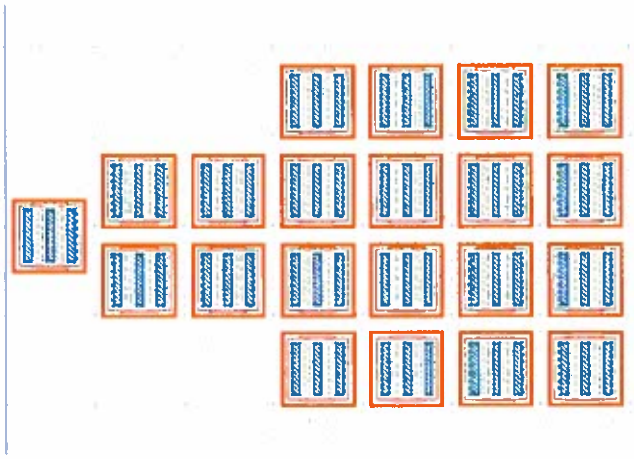


Figure 10 NM3 affected the simulation results, but could easily be lied out with NM0-1 current-mirror as a common centroid.

I continued with the assumption that this was to understand the limitations and trade-offs of the layout. Updating the layout from the schematic brought the components to what looked like a hint: I aimed to push the independent transistor *PM5* to the center where all the others could surround it symmetrically.



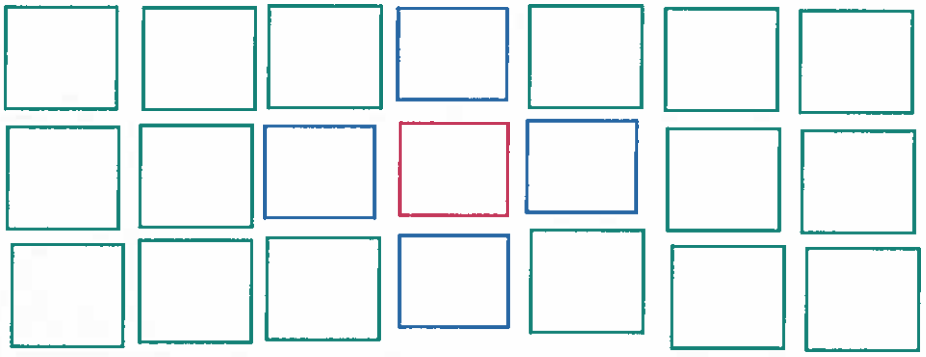


Figure 11 P5 is the further left individual capacitor, the next four to the right are P2, and lastly the remaining 16 are P4.

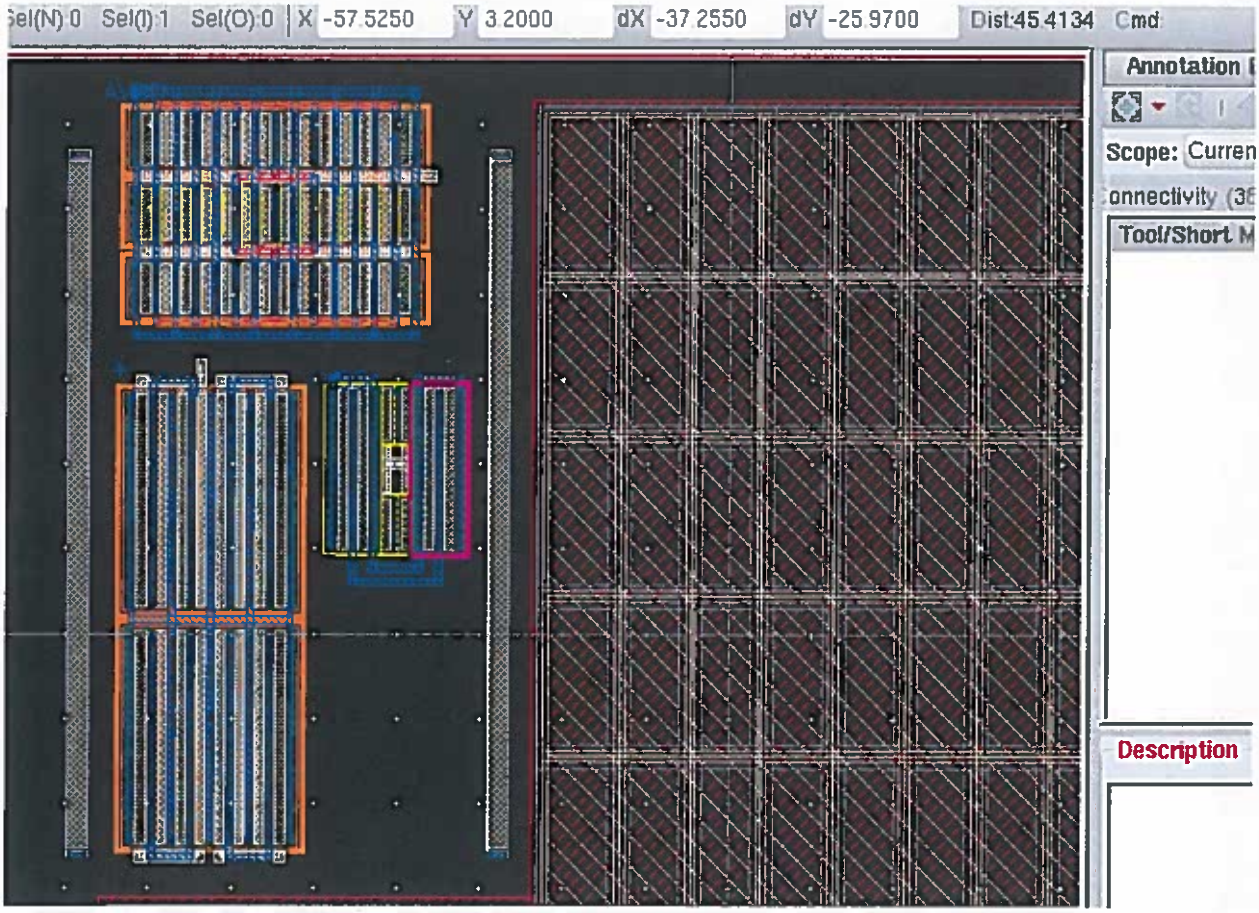


Figure 12 Near-final layout of the transistors in common-centroid configuration; top group is P2-4, left group is P0-1, and smallest on the right is N1-3. No guard rails, and can see parallel capacitors as well as the two resistors surrounding the transistors closest to their input and output.

The capacitors were a simple layout, although I had reduced their size by half and added doubled the amount; capacitors in parallel will sum, but it's important to note that their parasitics will not.

Lab 2 logbook submission_jkp16_attempt_2016-10-28-16-54-05
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Assignment: Lab 2 logbook submission
Date Submitted: Friday, 28 October 2016 16:54:05 o'clock BST
Current Grade: 45.5

Submission Field:
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Comments:
attached

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Name: Joshua Przepioski (jkp16)
Assignment: Lab 3 logbook submission
Date Submitted: Friday, 4 November 2016 16:17:00 o'clock GMT
Current Grade: 60.22

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